

Spectral-front identities for periodic Navier–Stokes and an obstruction to uniform pointwise Osgood closure

ns-singularity-certificate-lab*

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Abstract

For strong solutions of the unforced incompressible Navier–Stokes equations on $\mathbb{T}^3$ we prove that the spectral bandwidth $N_0^2 = \|\nabla u\|_2^2/\|u\|_2^2$ obeys the differential inequality $\frac{d}{dt} \log N_0^2 \leq KD/(2\nu)$, where $D = \|\nabla u\|_2^2$ and $K = \|\mathbb{P}(u \cdot \nabla u)\|_2^2/\|\nabla u\|_2^4$, with an explicit nonnegative defect; consequently if $\int_0^{T_{\max}} KD dt < \infty$ the solution is global, and in particular any finite-time breakdown forces the action to diverge. The action admits the exact representation $\int_0^{T'} KD dt = \int_0^{T'} \|\partial_t u\|_2^2/D dt + \nu^2 \int_0^{T'} \|\Delta u\|_2^2/D dt + \nu \log(D(T')/D(0))$; by AM–GM this yields the one-sided criterion $\int KD \leq 2 \int (\|\partial_t u\|_2^2/D + \nu^2 \|\Delta u\|_2^2/D)$ (the converse comparison fails for explicit decaying solutions), and KD is dominated pointwise by both the Serrin $(\infty, 2)$ and the critical $\|\nabla u\|_{L^3}^2$ actions. On the obstruction side we construct an explicit family of divergence-free trigonometric fields with exact closed-form spectral laws, and prove for it the capacity lower bound $\|\mathbb{P}(u_N \cdot \nabla u_N)\|_2^2 \geq c_0 N^3$ with the sharp exponent, uniformly in the cutoff profile and in the seed vector. It follows *unconditionally* that no Osgood-admissible pointwise field inequality $K \leq \Phi(\log N_0^2)$ can hold uniformly: the dichotomy cannot be upgraded to unconditional regularity through that route. The capacity constant c_0 and the band threshold N_* are the only non-effective constants in the paper; every other constant is explicit. Proofs are elementary and self-contained modulo classical local theory; no numerical statement enters any proof, and all numerics are confined to a certificate appendix. The claimed novelty is the exact packaging — the modal identities, their machine-checkable defect ledger, and the obstruction family with its capacity bound — and overlap with known refinements of the Serrin criteria is possible. Internal review was performed by AI agents and is not human peer review.

1 Introduction

The regularity theory of the three-dimensional Navier–Stokes equations is governed by critical quantities whose a-priori control is unknown. We isolate a scalar action, $\int KD dt$, that (i) bounds the growth of the spectral bandwidth through an exact modal identity with computable defects, (ii) is dominated by the classical critical actions, so its finiteness is a weaker hypothesis than theirs, and (iii) possesses an exact dynamic representation splitting it into a supercritical part (the $\dot{H}^1$ -bandwidth action) and a harmless logarithm. We then show, *unconditionally*, that the hypothesis cannot be verified by any pointwise Osgood-type field inequality: an explicit

*Machine-assisted research repository; all exact-arithmetic certificates cited below are reproducible from the repository at the commit recorded in the appendix. This draft claims no resolution of any Clay Millennium problem, in either direction.

smoothly truncated coherent critical-spectrum family obeys an exact capacity lower bound with the sharp exponent, which forces any uniform majorant $\Phi(\log N_0^2)$ of K to be of exponential class. The pair of results freezes this lane: the dichotomy is at least as strong as the classical criteria it refines, and the pointwise-Osgood route to making it unconditional is closed.

We emphasise what is *not* claimed: no global regularity for all data, no blow-up, no Clay statement. Numerics never enter any proof; they appear only as reproducible exact-arithmetic certificates corroborating the proven statements.

2 Setting and main results

Let $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ with normalised measure, $u(x, t) = \sum_{k \in \mathbb{Z}^3 \setminus \{0\}} \hat{u}_k(t) e^{ik \cdot x}$ real, zero-mean, divergence-free ($k \cdot \hat{u}_k = 0$), solving $\partial_t u + \mathbb{P}(u \cdot \nabla u) = \nu \Delta u$ with $\nu > 0$ and $u_0 \in H_\sigma^m$, $m > 5/2$, $u_0 \neq 0$; let u be the maximal strong solution on $[0, T_{\max})$. Write $H_r = \sum_k |k|^{2r} |\hat{u}_k|^2$, $D = H_1$, $N_0^2 = H_1/H_0$, $N_1^2 = H_2/H_1$, $z = \log N_0^2 (\geq 0)$, $\mathcal{N} = -\mathbb{P}(u \cdot \nabla u)$, and $K = \|\mathcal{N}\|_2^2/D^2$.

Theorem 1. *With the notation above:*

(a) $\Lambda(t) = \log N_0^2(t) - \frac{1}{2\nu} \int_0^t KD \, ds$ is non-increasing on $[0, T_{\max})$; quantitatively $\frac{d}{dt} \log N_0^2 = \frac{2\text{Cov}}{N_0^2} - \frac{2\nu V}{N_0^2} \leq \frac{KD}{2\nu}$ with $\text{Cov} = T_1/H_0$, $V = \text{Var}_p(|k|^2)$, $p(k) = |\hat{u}_k|^2/H_0$, and an explicit nonnegative two-part defect.

(b) Exactly one of: $\int_0^{T_{\max}} KD \, dt < \infty$, and then z is bounded, $u \in L^\infty(0, T_{\max}; H^1)$ and $T_{\max} = \infty$; or $\int_0^{T_{\max}} KD \, dt = \infty$. In particular $T_{\max} < \infty \Rightarrow \int_0^{T_{\max}} KD \, dt = \infty$; no claim is made that a global solution has finite action.

(c) For every $T' < T_{\max}$,

$$\int_0^{T'} KD \, dt = \int_0^{T'} \frac{\|\partial_t u\|_2^2}{D} \, dt + \nu^2 \int_0^{T'} N_1^2 \, dt + \nu \log \frac{D(T')}{D(0)},$$

and by AM-GM $\int_0^{T'} KD \, dt \leq 2 \int_0^{T'} (\|\partial_t u\|_2^2/D + \nu^2 N_1^2) \, dt$; the reverse comparison fails ($u = e^{-\nu t}(0, 0, \cos x_1)$ has $\int KD = 0$ with divergent bandwidth action).

(d) Pointwise, $KD \leq \|u\|_{L^\infty}^2$ and $KD \leq C_S^2 \|\nabla u\|_{L^3}^2$.

(e) If moreover $KD \leq \Phi(z)D + R$ a.e. with $\Phi > 0$ nondecreasing, $\int_0^\infty ds/\Phi = \infty$, $R \geq 0$, $\int_0^{T_{\max}} R \, dt < \infty$, then the first alternative of (b) holds.

The proof (Section 3) is a chain of eight elementary lemmas; the only external inputs are the classical local theory, the subcritical Serrin class $L^\infty L^6$, and the mean-zero Sobolev embedding.

On the obstruction side, we use a *smoothly* truncated coherent family.

Definition 1 (admissible cutoff). $\chi \in C^\infty([0, \infty); [0, 1])$ with $\chi \equiv 1$ on $[0, \frac{1}{2}]$ and $\text{supp } \chi \subset [0, 1]$; no monotonicity is assumed. Write $c_\chi = 1 + \|\chi'\|_\infty$.

Fix $v_0 \in \mathbb{R}^3 \setminus \{0\}$ (no integrality is needed) and an admissible χ , and let

$$\hat{u}_N(k) = \chi\left(\frac{|k|}{N}\right) \frac{P_k v_0}{|k|^2} \quad (k \neq 0), \quad \hat{u}_N(0) = 0, \quad P_k = I - \frac{k \otimes k}{|k|^2}, \quad (1)$$

a finite trigonometric polynomial banded at $|k| \leq N$.

Lemma 1 (exact family laws). u_N is real, zero-mean and divergence-free, and with $S_N^\chi = \sum_{k \neq 0} \chi(|k|/N)^2 |k|^{-2}$, $T_N^\chi = \sum_{k \neq 0} \chi(|k|/N)^2 |k|^{-4}$, $\Sigma_N^\chi = \sum_{k \neq 0} \chi(|k|/N) |k|^{-2}$: $H_0 = \frac{2}{3} \|v_0\|^2 T_N^\chi$, $H_1 = \frac{2}{3} \|v_0\|^2 S_N^\chi$, $N_0^2 = S_N^\chi / T_N^\chi$, $u_N(0) = \frac{2}{3} \Sigma_N^\chi v_0$, and $\|u_N\|_\infty^2 / \|\nabla u_N\|_2^2 \geq 2(\Sigma_N^\chi)^2 / (3S_N^\chi)$. Moreover, for $N \geq 32$ and every admissible χ ,

$$\frac{N}{348} \leq S_N^\chi \leq 128N, \quad 6 \leq T_N^\chi \leq 128, \quad \frac{N}{44544} \leq N_0^2 \leq \frac{64}{3}N,$$

all crude explicit constants, proven; so $N_0^2 \asymp N$.

The band-symmetry identity behind Lemma 1 is needed for a *general* radial weight f with $\sum |k|^2 |f| < \infty$ — no positivity and no sharp ball — since χ is not assumed monotone.

Theorem 2 (capacity lower bound; sharp exponent). *For every $v_0 \in \mathbb{R}^3 \setminus \{0\}$ and every admissible χ there exist $c_0 = c_0(\chi, v_0) > 0$ and $N_* = N_*(\chi, v_0)$ with*

$$\|\mathbb{P}(u_N \cdot \nabla u_N)\|_2^2 \geq c_0 N^3 \quad \text{for all } N \geq N_*.$$

The exponent 3 is sharp, since (d) with Lemma 1 gives $\|\mathbb{P}(u_N \cdot \nabla u_N)\|_2^2 \leq \|u_N\|_\infty^2 H_1 \asymp N^3$.

Theorem 2 replaces the capacity *hypothesis* of earlier versions of this paper. Its proof is given in full in the repository file `lstar/lstar_proof_main.md` (Theorem 7.1(3)) and is summarised in Section 3; we state here only what it delivers.

Theorem 3 (no Osgood-admissible uniform majorant). *If Φ is nondecreasing and $K(u) \leq \Phi(\log N_0^2(u))$ for every zero-mean divergence-free real trigonometric field u , then $\Phi(s) \geq ce^s$ for all large s ; in particular $\int^\infty ds/\Phi < \infty$, so clause (e) of Theorem 1 cannot be activated by any uniform pointwise field inequality.*

Remark 1 (what is open, and no longer used). The sharply truncated family $\hat{u}_N(k) = P_k v_0 / |k|^2$ on $1 \leq |k| \leq N$ satisfies the same exact laws with S_N, T_N in place of S_N^χ, T_N^χ and saturates the Bernstein ratio two-sidedly, $\frac{2}{3} S_N \leq \|u_N\|_\infty^2 / \|\nabla u_N\|_2^2 \leq S_N$. The corresponding capacity bound for that family — this paper’s former Hypothesis (L^*) — remains **open**. It is now **unused**: Theorem 3 quantifies over all real zero-mean divergence-free trigonometric fields, so the smooth family alone suffices. The honest statement is “the obstruction is unconditional”, not “(L^*) is proven”. A proven negative result constrains the sharp family further: no constant-vector sweeping split, in any Hölder pairing and any constant pairing direction, can exceed $\|\mathbb{P}(u_N \cdot \nabla u_N)\|_2^2 \gtrsim N$ — two powers short — which is why a concentrated test field, not a constant sweep, is the right instrument.

Remark 2 (effectivity). The constants c_0 and N_* of Theorem 2, and hence c of Theorem 3, are **non-effective**: the test field driving the capacity bound is produced from the density of $C_{c,\sigma}^\infty(\mathbb{R}^3)$ in $L_\sigma^2(\mathbb{R}^3)$, a pure existence argument. Every other constant in this paper is explicit. An effective version would require an explicitly constructed divergence-free Ψ with $\int (V \cdot \nabla V) \cdot \Psi \neq 0$ together with a quantitative lower bound on that pairing; non-vanishing is certified only on $\{0 < |\zeta| < \pi^2/2048\}$ intersected with two cone conditions, so such a Ψ has spatial radius $R \gtrsim 10^3$ and forces $N_* \gtrsim 10^4$.

3 Proofs

The full lemma chain L0–L11 with complete proofs is reproduced from the repository file `complete_proof.md`; we give it here in compressed form.

L0a (regularity of moments). H_0, H_1, H_2 are continuous; $H_0, H_1 \in C^1$ with $\dot{H}_0 = 2\langle u, \partial_t u \rangle$, $\dot{H}_1 = 2\langle -\Delta u, \partial_t u \rangle$ (difference quotients in $C^1 H^{m-2}$ paired against $C(L^2)$); $\partial_t u = \mathcal{N} - \nu(-\Delta)u$ in L^2 . Positivity of H_0 on all of $[0, T_{\max})$ is *not* elementary and is proven as L0b *after* L7, by bootstrap: if t_1 is the first vanishing time, then on $[0, t_1)$ Lemmas 1–4 and 7 apply, $KD \leq \|u\|_\infty^2 \leq C_m M^2$ with $M = \sup_{[0, t_1]} \|u\|_{H^m}$, so N_0^2 is bounded there, $\int_0^{t_1} N_0^2 < \infty$, and the integrated energy identity gives $H_0(t_1) > 0$, a contradiction.

L1 (ledger; energy neutrality). $\frac{1}{2}\dot{H}_r = T_r - \nu H_{r+1}$ ($r = 0, 1$) with $T_r = \sum |k|^{2r} \text{Re}\langle \hat{u}_k, \hat{\mathcal{N}}_k \rangle$; $T_0 = -\langle u, (u \cdot \nabla)u \rangle = 0$; hence $\int_0^{T_{\max}} D \, dt \leq H_0(0)/(2\nu)$.

L2 (identity). $\frac{d}{dt} \log N_0^2 = 2\text{Cov}/\mu - 2\nu V/\mu$ with $\mu = N_0^2$: subtract the logarithmic derivatives and use $H_0 H_2 - H_1^2 = H_0^2 V$.

L3 (closable covariance bound). $|\text{Cov}| \leq \sqrt{V\|\mathcal{N}\|_2^2/H_0}$: modal Cauchy–Schwarz $|\text{Re}\langle \hat{u}_k, \hat{\mathcal{N}}_k \rangle| \leq |\hat{u}_k| |\hat{\mathcal{N}}_k|$, then vector Cauchy–Schwarz.

L4 (square completion; (a)). $\sup_{V \geq 0} [\alpha\sqrt{V} - \nu V] = \alpha^2/(4\nu)$ with $\alpha = \sqrt{\|\mathcal{N}\|_2^2/H_0}$ gives $\frac{d}{dt} \log N_0^2 \leq \|\mathcal{N}\|_2^2/(2\nu H_1) = KD/(2\nu)$; the defect is $\Gamma^{\text{CS}} + \Gamma^{\text{SC}} \geq 0$ as displayed in the repository file, with rational total.

L5 (dichotomy; (b)). $\int KD < \infty \Rightarrow z \leq z_* \Rightarrow H_1 \leq e^{z_*} H_0(0) \Rightarrow u \in L^\infty L^6$ (mean-zero Sobolev), a subcritical Serrin class $\Rightarrow$ regularity and extension.

L6 (action representation; (c)). Expand $\|\mathcal{N}\|_2^2 = \|\partial_t u + \nu(-\Delta)u\|_2^2$, use $2\nu\langle \partial_t u, -\Delta u \rangle = \nu\dot{D}$, divide by $D > 0$, integrate; $\int \dot{D}/D = \log(D(T')/D(0))$.

L7 (dominations; (d)). $\|\mathcal{N}\|_2 \leq \|u\|_\infty \|\nabla u\|_2$ and $\|\mathcal{N}\|_2 \leq \|u\|_{L^6} \|\nabla u\|_{L^3} \leq C_S \|\nabla u\|_2 \|\nabla u\|_{L^3}$.

L8 (Osgood closure; (e)). $\Omega(z) = \int_0^z ds/\Phi$; $\frac{d}{dt}\Omega(z) \leq D/(2\nu) + R/(2\nu\Phi(0))$, both integrable; z bounded; apply L5.

L9–L11 and Theorem 3. Band symmetry $\sum_{k \neq 0} k_i k_j f = \delta_{ij} \frac{1}{3} \sum |k|^2 f$ for any radial f with $\sum |k|^2 |f| < \infty$ (sign flips kill $i \neq j$; permutations equalise the diagonal; no positivity is used); the family laws of Lemma 1 follow by applying it with the weights $\chi(|k|/N)^2 |k|^{-6}$, $\chi(|k|/N)^2 |k|^{-4}$ and $\chi(|k|/N) |k|^{-4}$ to $|P_k v_0|^2 = \|v_0\|^2 - (k \cdot v_0)^2/|k|^2$ and to $u_N(0) = \sum \hat{u}_N(k)$. The lattice bounds follow from the dyadic shell count $\#\{2^j \leq |k| < 2^{j+1}\} < 64 \cdot 8^j$ (upper) and an all-positive sub-box count in $N/2 < |k| \leq N$ (lower), using $\chi \equiv 1$ on $[0, \frac{1}{2}]$ for $S_N^\chi \geq S_{\lfloor N/2 \rfloor}$ and the six unit vectors for $T_N^\chi \geq 6$; the two-sided bound gives 2^q -adic gaps $c_- \leq s_{2^q N} - s_N \leq c_+$ with $q = 20$, $c_- = \log(1048576/950272) = 0.0985\dots$, $c_+ = \log(2^{20} \cdot 950272) = 27.63\dots$, elementary and N -uniform. By Theorem 2 and $H_1 \leq \frac{256}{3} \|v_0\|^2 N$, $K(u_{N_j}) \geq c_0'' e^{s_{N_j}}$ along $N_j = 2^{20j} N_\sharp$; monotonicity of Φ interpolates across the bounded gaps, so $\Phi(s) \geq c_0'' e^{-c_+ s}$ for all large s and $\int^\infty ds/\Phi < \infty$.

Theorem 2 (summary; full proof in the repository). Because $P_k = P_{k/N}$, the family obeys the exact scaling $\hat{u}_N(k) = N^{-2}F(k/N)$ with $F(\xi) = \chi(|\xi|)P_\xi v_0/|\xi|^2$, so with $V = \tilde{F}$ (inverse transform on $\mathbb{R}^3$) the lattice sums defining $u_N(y/N)$ and $\nabla u_N(y/N)$ are *literally* Riemann sums of spacing $1/N$: $u_N(y/N) = N(V(y) + E_N(y))$, $\nabla u_N(y/N) = N^2(\nabla V(y) + E'_N(y))$ identically, with $\sup_{|y| \leq R}(|E_N| + |E'_N|) = O(N^{-1} \log N)$ explicitly (the integrand is compactly supported with a single integrable algebraic singularity; the gradient half needs and receives its own, log-free, argument). Poisson summation is *not* available here — V decays exactly like $|y|^{-1}$ with a sign-definite spherical mean, so the periodisation diverges and, its terms being eventually positive, no summation method rescues it. Pairing against the concentrated divergence-free test field $\psi_N(x) = N^{3/2}\Psi(Nx)$, whose $L^2(\mathbb{T}^3)$ norm is N -independent, and using self-adjointness of $\mathbb{P}$ gives $\|\mathbb{P}(u_N \cdot \nabla u_N)\|_2^2 \geq (2\pi)^{-3}N^3|I_\Psi + \mathcal{E}_N|^2/\|\Psi\|_{L^2(\mathbb{R}^3)}^2$ with $I_\Psi = \int (V \cdot \nabla V) \cdot \Psi$ and $\mathcal{E}_N = O(\log N/N)$. It remains to find one Ψ with $I_\Psi \neq 0$, i.e. to show $\mathbb{P}(V \cdot \nabla V) \neq 0$. This holds for *every* admissible χ and *every* v_0 , because the leading small- ζ singularity of $F * F$ is the χ -independent Oseen convolution τ , for which, with $v_0 = \|v_0\|e_3$,

$$\zeta \times (\tau(\zeta)\zeta) = \frac{3\pi^3}{8}\|v_0\|^2 \frac{\zeta_3}{|\zeta|}(\zeta \times e_3) \neq 0$$

off the two degenerate directions — the Fourier form of the elementary fact that the Oseen field is not a stationary Euler flow, since $\nabla \times (W \times \nabla \times W) = 12y_3(e_3 \times y)/|y|^6$. A crude perturbation bound $|T - \tau| \leq 64\pi\|v_0\|^2$ on $|\zeta| \leq \frac{1}{8}$ transfers this to the cutoff family on $|\zeta| < \pi^2/2048$. $\square$

4 Position among known criteria; limitations

By (d), the finiteness of $\int KD$ is implied by the classical Serrin $(\infty, 2)$ action and by $\int \|\nabla u\|_{L^3}^2 dt$; the criterion (b) is therefore at least as strong as both (we do not claim strictly stronger). By (c) and its AM–GM corollary, the criterion is implied by the a-priori control of $\int (\|\partial_t u\|_2^2/D + \nu^2 N_1^2) dt$ — the half-derivative-supercritical $\dot{H}^1$ -bandwidth action — while the exact identity shows the two differ only by $\nu \log D$, which can diverge to $-\infty$ on decaying solutions; this makes precise both why the criterion is strong and why its unconditional verification is out of reach of the present methods. Theorem 3 closes the pointwise-Osgood route unconditionally. Its one limitation is effectivity (Remark 2): the capacity constant c_0 and the threshold N_* are not produced, only shown to exist. All statements are elementary; the contribution is the exact packaging, the obstruction family with its capacity bound, and the machine-verifiable defect ledger accompanying (a).

A Certificate appendix (numerics, separated from all proofs)

No number below enters any proof. Both Theorem 1 and Theorem 3 are unconditional, so — unlike in earlier versions — these values no longer carry a hypothesis; they are corroboration.

Smooth family (the family used by Theorem 3). Reproducible from the repository (experiments/run_smooth_family_capacity.py; JSON with SHA-256 sidecar at outputs/lstar/smooth_family). Seed $v_0 = (1, 2, 3)$; weight $\chi(r) = 1 - S(\text{clamp}((r^2 - \frac{1}{4})/\frac{3}{4}, 0, 1))$ with S the degree-9 smoothstep $126s^5 - 420s^6 + 540s^7 - 315s^8 + 70s^9$, which is a polynomial in r^2 — hence $\chi(|k|/N) \in \mathbb{Q}$ at every lattice point and the whole family is exactly rational — and lies in C^4 , not C^∞ . Definition 1

asks for C^∞ ; the proof uses smoothness of χ only through $c_\chi = 1 + \|\chi'\|_\infty$, so C^1 suffices there, but as the class is written the exact lanes below are evidence for the C^p variant. Degree- $(2p+1)$ smoothsteps give rational $\chi \in C^p$ for every p .

Exact-Fraction verification of the family laws at $N = 4, 6, 8, 10, 12$: every residual of H_0 , H_1 and $u_N(0)$ is the rational 0, with $u_N(0)$ exactly parallel to v_0 . Exact capacity by full $O(|B_N|^2)$ contraction over all ordered pairs of active band points:

N	4	6	8	10	12	16
$\ \mathbb{P}(u_N \cdot \nabla u_N)\ _2^2/N^3$	678.28	1027.31	1241.51	1384.36	1485.78	1619.65
K	0.6166	1.1359	1.6705	2.2077	2.7464	3.8254
K/N_0^2	0.2662	0.3287	0.3643	0.3861	0.4008	0.4195

A dealiased FFT continuation (grid $4N+2$, whole product spectrum represented) extends this to $N = 48$: $\|\mathbb{P}(u_N \cdot \nabla u_N)\|_2^2/N^3$ rises $1241.5 \rightarrow 1911.3$ and K/N_0^2 rises $0.3643 \rightarrow 0.4574$ over $N = 8 \rightarrow 48$, both monotonically. Exact versus FFT at the two shared bands agrees to double-precision roundoff: relative difference 5.5×10^{-16} ($N = 8$) and 7.0×10^{-16} ($N = 16$) in the norm, 5.3×10^{-16} and 8.1×10^{-16} in K . Two-point Richardson diagnostics (not proofs): $\|\mathbb{P}(u_N \cdot \nabla u_N)\|_2^2/N^3 \rightarrow 2064.3$, $K/N_0^2 \rightarrow 0.4764$, $H_1/N \rightarrow 87.51$, $N_0^2/N \rightarrow 0.5671$ — positive limits, i.e. the sharp exponent, consistent with Theorem 2. The continuum non-degeneracy $\mathbb{P}(V \cdot \nabla V) \not\equiv 0$ is corroborated three independent ways (exact rational lattice modes; float 3-D quadrature of the continuum convolution; an axisymmetric radial reduction giving $\text{curl}(V \cdot \nabla V) = Z(r) \sin \theta \cos \theta e_\phi$ with $Z \rightarrow -12\pi^4 \|v_0\|^2/r^4$), agreeing to 10^{-4} or better.

Sharp family (corroboration only; family no longer used). These are *sharp*-family numbers, supporting the open Hypothesis (L^*) of Remark 1, which no statement in this paper uses. Reproducible from `experiments/run_coherent_family_certificates.py` and `run_osgood_gate.py` (JSON with SHA-256 sidecars under `outputs/verification_sprint_v1/osgood_gate/`): exact-Fraction family laws at $N = 4, 6, 8$ (residual 0); exact $K(u_4) = 0.78841070\dots$ (full fraction stored), $K(u_6) = 1.43447181\dots$, $K(u_8) = 2.03724301\dots$; float pipeline agreement 0.0, 4.6×10^{-16} , 1.3×10^{-15} ; $Q(v_0) > 0$ exact at $N = 4, 6, 8$ with float continuation $Q/N^2 = 850.7 \rightarrow 1029.5$ over $N = 8 \rightarrow 32$; captured fraction of $\|\mathbb{P}(u \cdot \nabla u)\|_2$ by the sweeping pairing: 0.921/0.917/0.915 (exact, $N = 4, 6, 8$) and $0.915 \rightarrow 0.912$ (float, $N = 8 \rightarrow 32$); $K/N_0^2 = 0.259 \rightarrow 0.396$ ($N = 4 \rightarrow 32$); Leray retention $0.780 \rightarrow 0.867$.